
UNIT 10 MULTIVARIABLE REGRESSION MODELS

Structure

- 10.0 Objectives
- 10.1 Introduction
- 10.2 Regression Model with Two Explanatory Variables
 - 10.2.1 Estimation of Parameters
 - 10.2.2 Variance and Standard Errors
- 10.3 Interpretation of Regression Coefficients
 - 10.3.1 Goodness of Fit: Multiple Coefficient of Determination: R^2
 - 10.3.2 Analysis of Variance (ANOVA)
- 10.4 Inclusion and Exclusion of Variables
- 10.5 Generalisation to n-explanatory Variables
- 10.6 Problem of Multi-co-linearity
- 10.7 Problem of Hetero-scedasticity
- 10.8 Problem of Autocorrelation
- 10.9 Maximum Likelihood Estimations
- 10.10 Let Us Sum Up
- 10.11 Key Words
- 10.12 Answers or Hints for Check Your Progress Exercises

10.0 OBJECTIVES

This Unit aims at equipping the learners with techniques of multiple regressions which come handy in analysing a number of situations where one is required to study impact of a number of independent variables (factors) on some particular dependent variable. After going through this unit, you will be able to:

- have a fairly good comprehension of technique of multiple regression analysis;
- appreciate the need for extending the basic idea of regression as discussed in Unit 8;
- apply those techniques for more realistic model formulation of explanation of economic phenomena; and
- spot and avoid pit falls which may arise (in quantitative analysis).

on accounts of auto-correlation, hetero-scedasticity and multi-co-linearity on the one hand and mis-specification of the model, (in the sense of including irrelevant as well as excluding relevant variables from the model) on the other.

10.1 INTRODUCTION

In Unit 8, you have studied basics of the Classical Linear Regression Model. You regressed dependent variable Y on independent variable X. In other

words, you tried to explain changes in Y in terms of the changes in X . You had hypothesised a linear relationship between Y and X of this type:

$$Y = \alpha + \beta X$$

Then, you went on to 'estimate' the two constants $\hat{\alpha}$ and $\hat{\beta}$. Once the estimates were ready, you had the estimated model or 'relationship' with you given by

$$\hat{Y} = \hat{\alpha} + \hat{\beta} x$$

In the present unit we are going to extend this type of analysis further to make it more 'realistic' and 'comprehensive'. We will do it in a number of steps: first of all, we shall introduce one more explanatory variable and re-examine the model. Next step will be to generalise the model to n -explanatory variables. At the next stage, we try to interpret the partial regression coefficients. After that, we will try to examine how large should the number ' n ' be – that is how many explanatory variables must be included in the model and what should be the statistical touch stone for arriving at such a decision. Then, we examine the conditions, or 'assumptions', which make these extensions and generalisations possible. We also examine the possible effects of violation of one or more assumptions. We shall, in particular, pay attention to problems of **multi-co-linearity**, **Hetero-scedasticity** and **auto-correlation**. Then, we take a look at another class of estimates, the Maximum Likelihood Estimators. Finally, we give a brief exposition to some uses of multiple regression analysis in economics.

10.2 REGRESSION MODEL WITH TWO EXPLANATORY VARIABLES

At the outset, please note one small change in notation: in section 10.1 – the introduction to the present unit, we have used the specification of model as

$$Y = \alpha + \beta X$$

However, now onwards, we will write the model in slightly different form:

$Y = \beta_0 + \beta_1 X_1$ This form helps us in presentation in the sense that as and when, we add more explanatory variables X_2, X_3 , etc. we simply indicate their coefficients with respective $\beta_1, \beta_2, \beta_3$, etc.

So the model that we consider in the present section has two explanatory variables X_1 , and X_2 . The non-stochastic specification will be:

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \quad (10.1)$$

The same model in stochastic form will have 'error terms' included:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \mu \quad (10.2)$$

$$= E(Y) + \mu \quad (10.3)$$

We can use subscripts ' t ' with Y_0, X_1, X_2 and μ to denote ' t^{th} ' observation. Thus, the above equations can be written as

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t}$$

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \mu_t \text{ and}$$

$$Y_t = E(Y_t) + \mu_t \text{ respectively}$$

In the relationships above, the component $(\beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t})$ is systematic or deterministic component, which is equal to mean value $E(Y_t)$ — or a point on the regression line.

The component μ_t is non-systematic random component determined by factors other than X_1 and X_2 .

The model specified by equations 10.1 and 10.2 is a linear regression model which is linear in parameters.

10.2.1 Estimation of Parameters: The Ordinary Least Squares Approach

The Ordinary Least Squares Approach

We collect a sample of observations on Y_1 , X_1 and X_2 and write down sample regression function

$$Y_t = b_0 + b_1 X_{1t} + b_2 X_{2t} + e_t \quad (10.4)$$

Note that b_0 , b_1 , b_2 have replaced the corresponding population parameters β_1 , β_2 , β_3 here. The random component μ_t replaced by e_t or the sample error term.

So, simply speaking:

b_0 = the estimator for β_0

b_1 = the estimator for β_1

b_2 = the estimator for β_2

We know from the Unit 8 that the *principle of ordinary least squares* selects those values for unknowns (b_0, b_1, b_2) such that residual sum of squares (RSS) $= \sum e_t^2$ is minimum. We can develop this idea by rewriting equation 10.4 as

$$e_t = Y_t - b_0 - b_1 X_{1t} - b_2 X_{2t} \quad (10.5)$$

Square on both sides and sum up to get RSS:

$$\sum e_t^2 = \sum [Y_t - b_0 - b_1 X_{1t} - b_2 X_{2t}]^2 \quad (10.6)$$

Differentiating the (10.6) w.r.t b_0 , b_1 , b_2 and equating to zero gives us the three normal equations:

$$\bar{Y}_t = b_0 + b_1 \bar{X}_{1t} + b_2 \bar{X}_{2t} \quad (10.7)$$

$$\sum Y_t X_{1t} = b_0 \sum X_{1t} + b_1 \sum X_{1t}^2 + b_2 \sum X_{1t} X_{2t} \quad (10.8)$$

$$\sum Y_t X_{2t} = b_0 \sum X_{2t} + b_1 \sum X_{1t} X_{2t} + b_2 \sum X_{2t}^2 \quad (10.9)$$

These three equations give us the following expressions for b_0 , b_1 , and b_2 respectively:

$$b_0 = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2 \quad (10.10)$$

$$b_1 = \frac{(\sum y_t x_{1t})(\sum x_{2t}^2) - (\sum y_t x_{2t})(\sum x_{1t} x_{2t})}{(\sum x_{1t}^2)(\sum x_{2t}^2) - (\sum x_{1t} x_{2t})^2} \quad (10.11)$$

and

$$b_2 = \frac{(\sum y_t x_{2t})(\sum x_{1t}^2) - (\sum y_t x_{1t})(\sum x_{1t} x_{2t})}{(\sum x_{1t}^2)(\sum x_{2t}^2) - (\sum x_{1t} x_{2t})^2} \quad (10.12)$$

The lower case letters denote, as usual, the deviations from the respective means. Thus $y_t = (y_t - \bar{Y})$, $x_{1t} = (X_{1t} - \bar{X}_1)$ and $x_{2t} = (X_{2t} - \bar{X}_2)$..

10.2.2 Variance and Standard Errors

The variances of OLS estimators given in 10.10, 10.11 and 10.12 above are given in terms of means and deviations of X_1 and X_2 and the variance of population error terms μ_t . Thus,

$$Var(b_0) = \left(\frac{1}{n} + \frac{\bar{X}_1^2 \sum x_{2t}^2 + \bar{X}_2^2 \sum x_{1t}^2 - 2 \bar{X}_1 \bar{X}_2 \sum x_{1t} x_{2t}}{\sum x_{1t}^2 \sum x_{2t}^2 - (\sum x_{1t} x_{2t})^2} \right) \sigma^2 \quad (10.13)$$

$$\text{Standard Error of } b_0 = \sqrt{Var(b_0)} \quad (10.14)$$

$$Var(b_1) = \frac{\sum x_{2t}^2}{\sum x_{1t}^2 \sum x_{2t}^2 - (\sum x_{1t} x_{2t})^2} \cdot \sigma^2 \quad (10.15)$$

$$SE(b_1) = \sqrt{Var(b_1)} \quad (10.16)$$

and

$$Var(b_2) = \frac{\sum x_{1t}^2}{\sum x_{1t}^2 \sum x_{2t}^2 - (\sum x_{1t} x_{2t})^2} \cdot \sigma^2 \quad (10.17)$$

$$SE(b_2) = \sqrt{Var(b_2)} \quad (10.18)$$

When σ^2 is unknown and its unbiased OLS estimator is obtained, we find that

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{n-3} \quad (10.19)$$

The denominator of 10.19 shows the degrees of freedom. In a sample of size 'n' we exhaust 3 degrees of freedom in estimating b_0 , b_1 , and b_2 . Note that this reasoning is quite general, in the sense that if out of a sample of size 'n', we

estimate 'k' parameters, remaining degrees of freedom will be $n - k$. We call $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$, the standard error of estimate/regression.

The residual sum of squares in 10.19 is $\sum e_t^2 = \sum (Y_t - \hat{Y}_t)^2$. This expression is equivalent to the following.

$$\sum e_t^2 = \sum y_t^2 - b_1 \sum y_t x_{1t} - b_2 \sum y_t x_{2t} \quad (10.20)$$

10.3 INTERPRETATION OF REGRESSION COEFFICIENTS

Mathematically, b_1 and b_2 represent the partial slopes of regression plan with respect to X_1 and X_2 respectively. In other words, b_1 shows the rate of change in Y as X_1 alone undergoes a unit change, keeping all other thing, constant. Similarly, b_2 represents rate of change of Y as X_2 alone changes by a unit while other things are held constant.

Note: This interpretation is quite similar to what you have been doing in economic theory. Recall the law of demand: quantity demanded of a commodity varied inversely with its price, 'ceteris paribus' that is when other things were kept 'constant'. What were those other things? Those were prices of complements and substitutes, income of the consumer, and tastes/likings/disliking etc.

10.3.1 Goodness of Fit: Multiple Coefficient of Determination: R^2

Recall that in case of a single independent variable, r^2 measured goodness of fit of the fitted sample regression line. It gave you the percentage total variation in dependent variable Y which has been explained by the single explanatory variable X . Correspondingly, when we have two explanatory variables X_1 and X_2 , we would like to know the proportion of total variation in $Y = \sum y_t^2$ explained by X_1 and X_2 jointly. This information is conveyed by R^2 — the multiple coefficient of determination. Thus,

$$R^2 = \frac{ESS}{TSS}$$

When ESS is explained sum of squares and TSS is total sum of squares. You are familiar with the relationship between TSS, ESS and RSS from Unit 8.

You know that $ESS = b_1 \sum y_t x_{1t} + b_2 \sum y_t x_{2t}$

and $RSS = \sum y_t^2 - b_1 \sum y_t x_{1t} - b_2 \sum y_t x_{2t}$

Therefore R^2 can be computed as

$$R^2 = \frac{b_1 \sum y_t x_{1t} + b_2 \sum y_t x_{2t}}{\sum y_t^2} \quad (10.21)$$

You can get the quantities in the above formula from your computation sheet for the normal equations. The R^2 lies between 0 and 1. Closer it is to one;

better is the fit – implying estimated regression line is capable of explaining greater proportions of variation in Y. The positive square root of R^2 is called coefficient of multiple correlation.

10.3.2 Analysis of Variance (ANOVA)

We know the relationship:

$$TSS = ESS + RSS$$

This is equivalent to saying:

$$\sum y_i^2 = b_1 \sum y_i x_{1i} + b_2 \sum y_i x_{2i} + \sum e_i^2$$

$$(TSS) = (ESS) + (RSS)$$

A study of the components of TSS is called analysis of variance in the context of regression. You know that every sum of squares has some degrees of freedom (df) associated with it. In our above 2-explanatory variable case, the degrees of freedom will be

$$TSS = n-1$$

$$RSS = n-3$$

$$ESS = 2$$

We can put this information in form of a table:

Table 10.1: ANOVA Table for 2-Explanatory Variable Regression

Source of Variation	Sum of Squares (SS)	df	Mean S.S = $\frac{SS}{df}$
Due to regression (ESS)	$b_1 \sum y_i x_{1i} + b_2 \sum y_i x_{2i}$	2	$\frac{ESS}{2}$
Due to residuals	$\sum e_i^2$	n-3	$\frac{\sum e_i^2}{n-3}$
TSS	$\sum y_i^2$	n-1	

One may be interested in testing a null hypothesis $H_0: B_1 = B_2 = 0$. In such a case we find that

$$\frac{ESS / df}{RSS / df}$$

that is ratio of the variance explained by X_1 and X_2 to unexplained

variance follows F distribution with 2 and n-3 degrees of freedom. In general, if regression equation estimates 'k' parameters including intercept, then F ratio has (k-1) df in numerator and (n-k) df in the denominator.

Interpretation: Larger the variance explained by fitted regression line, larger the numerator will be in relation to denominator. Thus a larger F value is evidence **against** the 'truthfulness' of $H_0: B_1 = B_2 = 0$. That is, F values larger than one will indicate that hypothesis of both the variables X_1 and X_2 having no effect on Y cannot sustain.

We can also express F values in terms of R^2 .

$$F = \frac{R^2 / (k - 1)}{(1 - R^2) / (n - k)}$$

Check Your Progress 1

1) What is multiple regression? Explain with an example.

.....

.....

.....

.....

.....

2) State the assumptions of classical linear multiple regression model.

.....

.....

.....

.....

.....

3) What do you mean by linearity of regression model? Explain?

.....

.....

.....

.....

.....

4) How do you interpret coefficients of multiple regression model?

.....

.....

.....

.....

.....

.....

10.4 INCLUSION AND EXCLUSION OF EXPLANATORY VARIABLES

The Adjusted $R^2 : \bar{R}^2$

It has been noted that as we add more and more explanatory variables X_s , the explained sum of squares keeps on rising. Thus, R^2 goes on increasing as we increase X_s , the explained sum of squares keeps on rising. Thus, R^2 goes on increasing as we increase X_s . But, notice that adding each additional variable 'eats up' one degree of freedom and our definition of R^2 makes no allowance for loss of degrees of freedom. Hence, the thinking that you can improve the goodness of fit by suitably increasing the number of variables may not be justified — We know that TSS always has $(n-1)$ degrees of freedom. Therefore, comparing two regression models with same dependent variable but differencing number of independent variables will not be justified. Therefore, we must adjust our measure of goodness of fit for degrees of freedom. This measure is called adjusted R^2 , denoted by $\bar{R}^2$. It can be derived from R^2 as follows:

$$\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)} \quad (10.22)$$

Therefore, it is recommended that, one must include new variables only if (upon inclusion) $\bar{R}^2$ increases and not otherwise. A general guide is provided by 't' statistic, if absolute value of the co-efficient of added variable is greater than one, retain it (Note that 't' value is calculated under the hypothesis that population value of that co-efficient is zero).

10.5 GENERALISATION TO N-EXPLANATORY VARIABLES

In general, our regression model may have a large number of independent variables. Each of those variables can, on priority grounds, be expected to have some influence over the 'dependent' or 'explained' variable. Consider a very simple example. What can be determinants of demand for potatoes in a vegetables market? One obvious choice will be the price of potatoes. What else can affect the quantity demanded? Could it be availability of vegetables which can be paired off with potatoes? In that case, prior of a large number of vegetables which are cooked along with potatoes will become 'relevant explanatory variables'. You cannot ignore income of the community that patronizes the particular market. Needless to say, the dietary preferences of members of the households can also affect the demand and so on. In the next Unit, we shall discuss techniques which help us restrict the analysis to a selected 'few variables, though theoretic considerations may find a huge number of them to be 'useful' and 'powerful' determinants. In fact, in economic theory, we usually append the phrase **Ceteris paribus**, with many a statements. This phrase means keeping all other things constant. That means, we may focus on impact of only a few selected variables on the dependent variable while assuming that all other variables remain 'unchanged' during the period of analysis. This assumption may not hold so, we have to juggle the need to include more and more variables in our model with the 'gains' in

explanatory power of the model. We have developed, in previous section (10.4) a working touchstone for inclusion of more variables in terms of improvement in $\overline{R^2}$ and have tried to give it 'practical' shape in form of magnitude of 't' values of the relevant slope parameters.

With these considerations in mind we can generalise the linear regression model as follows:

We hypothesize that in population, the dependent variable Y depends upon n explanatory variables, $X_1, X_2, \dots, X_n$. We also assume that the relationship is linear in parameters. Three more assumptions are made and they have very significant bearing on the analysis. These are:

- a) Absence of Multi-co-linearity;
- b) Absence of Hetero-scedasticity; and
- c) Absence of Autocorrelation

We will discuss the complications, which arise because of violations of these assumptions in section 10.6, 10.7 and 10.8 respectively. So our Classical Linear General Regression Model is :

$$Y_t = \beta_0 + \beta_1 X_{1t} + \dots + \beta_n X_{nt}$$

or

where it is understood that Y and each X will have a large numbers of values $t=1, \dots, N$, forming (n+1) 'tuples' ($Y_{11}, X_{11}, X_{21}, \dots, X_{n1}$).

We can simply write

$$Y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \mu \quad (10.23)$$

in Matrix equation form, we get

$$Y = X\beta + U \quad (10.24)$$

Where $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$ and $U = \begin{bmatrix} U_1 \\ \vdots \\ U_n \end{bmatrix}$

also $X = \begin{bmatrix} X_{11}, X_{21}, X_{31}, \dots, X_{k1} \\ X_{12}, X_{22}, X_{32}, \dots, X_{k2} \\ \vdots \\ X_{1n}, X_{2n}, X_{3n}, \dots, X_{kn} \end{bmatrix}$

We assume that (1) expected values of error terms are equal to zero; that is $E(u_i) = 0$ for all 'i'. In matrix notation

$$E(u) = \begin{bmatrix} E(u_1) \\ \vdots \\ E(u_n) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0$$

- 2) The error terms are not correlated with one another and they all have same variance for σ^2 all sets values of the variables X. That is,

$$E(u_i u_j) = 0; \quad \forall i \neq j \text{ and}$$

$$E(u_i^2) = \sigma^2 \forall i$$

in matrix notation:

$$E[UU'] = \begin{bmatrix} E(u_1^2), E(u_1 u_2) \dots E(u_1 u_n) \\ E(u_1 u_2), E(u_2^2) \dots E(u_2 u_n) \\ \vdots \\ E(u_1 u_n), E(u_2 u_n) \dots E(u_n^2) \end{bmatrix} = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 \\ 0 & \sigma^2 & \sigma & 0 \\ 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & \sigma^2 \end{bmatrix} = \sigma^2 I_n$$

- 3) Number $X_{1i} \dots X_{ki}$ are all real numbers and are free of random errors or disturbances.
- 4) The matrix X has been linearly independent columns. It implies that number of observations exceeds number of co-efficient to be estimated. It also implies there exists no exact linear relationships between any of the X variables.

Note: Assumptions that $E(U_i U_j) = 0$ means that error terms are not correlated. The implication of **diagonal** terms in matrix $E(UU')$ being all equal to σ^2 is that all error terms have same variance, σ^2 . This is also called assumption of homo-scedasticity. The last assumption implies absence of multi-co-linearity. We can write the regression relation for the sample as:

$$e = Y - Xb$$

where e , Y , X and b are appropriate matrices.

Sum of squared residuals will be

$$\begin{aligned} \phi &= \sum e_i^2 = \sum (Y_i - b_1 X_{1i} \dots + b_k X_{ki})^2 \\ &= e'e = [Y - Xb]' [Y - Xb] \\ &= Y'Y - 2b'X'Y + b'X'Xb \end{aligned} \quad (10.25)$$

Note: $b'X'Y$ is scalar and is therefore equal to its transpose $Y'Xb$.

by equating 1st order partials of ϕ w.r.t each b_i , to zero, we get k normal equations. This set of equations in matrix form is:

$$\frac{\partial \phi}{\partial b} = -2X'Y + 2X'Xb = 0 \quad (10.26)$$

$$X'Xb = X'Y \quad (10.27)$$

when X has rank equal to k , the normal equation 10.27 will have a unique solution and least squares estimator b is equal to:

$$b = [X'X]^{-1}[X'Y] \quad (10.28)$$

We have assumed b to be estimator for β and thus $E(b) = \beta$, therefore we can rewrite 10.28 as

$$\begin{aligned} b &= [X'X]^{-1}X'[X\beta + u] \\ &= [X'X]^{-1}X'X\beta + [X'X]^{-1}X'U \\ &= \beta + [X'X]^{-1}X'U \\ \therefore E(b) &= E(\beta) + E[[X'X]^{-1}X'E(u)] = E(\beta) + [X'X]^{-1}X'E(u) \\ &= \beta \end{aligned}$$

$$\text{Variance of } b = \sigma^2(X'X)^{-1}$$

Notes

- 1) In this course our objective is simply to introduce the concepts. Those who plan to pursue the concepts at much more rigorous level can study our course on Econometric Methods (MEC) – included as optional course of M.A.(Economics) Programme.
- 2) The other ideas regarding coefficient of determination R^2 and adjusted R^2 remain the same as they were developed for two explanatory variable case.

Now we can safely turn to discussions about non-satisfaction or violation of assumptions.

10.6 THE PROBLEM OF MULTI-CO-LINEARITY

Many a times our X variables may be found to have some other linear relationships among themselves. This vitiates our classical regression model.

Let us illustrate it with help of our 2 explanatory variable model.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + U_i$$

Let us give specific names to variables, say, X_1 is price of commodity Y and X_2 is family income. We expect β_1 to be negative and β_2 to be positive. Now we go one step further. Let Y be demand for milk, X_1 be price of milk and let the family wise demand for milk is being estimated for a family, which produces and sells milk! Clearly, larger the value of X_1 higher the magnitude of X_2 will be.

In such situations, the estimation of co-efficient will not be possible. Recall, we wanted variables X in our matrix equations to be linearly independent. If that condition is not satisfied X matrix becomes singular, that is its determinant tends to vanish. Thus, there will be **no solution** to the normal equation **10.26 (or 10.27)**.

However, if co-linearity not perfect, we can still get OLS estimates and they remain best linear unbiased estimates (BLUE) – though one or more partial regression co-efficient may turn out to be individually insignificant.

Not only this, the OLS estimates still retain property of minimum variance. Further, it is found that multi co-linearity is essentially a sample regression problem. The X variables may not be linearly related in population but some of our suppositions while drawing a sample may create a situation of multiple linear relations.

The practical consequences of multi-co-linearity. Gujarali, (D.N.) has listed the following consequences of multiplicity of linear relationships:

- 1) Large variances /SEs of OLS estimates
- 2) Wider confidence intervals
- 3) Insignificant 't' ratios for β parameters
- 4) A high R^2 despite few significant t values
- 5) Instability of OLS estimators: The estimators and their standard errors (SEs) become very sensitive to small changes in data.
- 6) Sometimes, even signs of some of the regressions may turn out to be theoretically unacceptable like a rise in income having negative impact on demand for milk.
- 7) When many regressions have insignificant coefficients, their individual contributions to the explained sum of squares cannot be assessed properly.

The multi-co-linearity can be detected by:

- 1) high R^2 but few significant 't' ratios,
- 2) high pair wise correlation between explanatory variables. One can try partial correlations, subsidiary or auxiliary regressions as well. But each such technique increases burden of calculations.

Check Your Progress 2

- 1) When do you decide to drop a variable from the regression equation? Why?

.....

.....

.....

.....

- 2) When do you include more variable(s) into your model? Why?

.....

.....

.....

.....

- 3) “Inclusion of more variables always increases R^2 the goodness of fit. So to make a regression model ‘good’ what we need to do is simply increase the number of explanatory variables”. Do you agree/disagree with this statement? Give reasons.

.....

.....

.....

.....

.....

- 4) What is multi-co-linearity? What are its consequences?

.....

.....

.....

.....

10.7 PROBLEM OF HETERO-SCEDASTICITY

The Classical Linear Regression Model has a significant underlying assumption in homo scedasticity, that is, all the error terms are identically distributed with mean equal to zero and standard deviation equal to σ (or variance equal to σ^2). σ^2 What happens when this second part of assumption regarding distribution of variance does not hold? As a result, in symbolic terms $E(u_i)^2 = \sigma_i^2$, that is, if the expectation of squared errors is no longer equal to σ^2 — each error term has its own σ^2 , or variance varies from observation to observation.

It has been observed that usually time series data does not suffer from this problem of hetero scedasticity but in cross-section data, the problem may assume serious dimensions. **The consequences of hetero scedasticity:** If the assumption of homoscedasticity does not hold, we observe the following impact on OLS estimators.

- 1) They are still linear
- 2) They are still unbiased
- 3) But they no longer have minimum variance – that is we cannot call them BLUE: the Best Linear Unbiased Estimators. In fact, this point is relevant both for small as well as large samples.
- 4) Reason for this problem hinted at in (3) above is that generally, OLS estimators have some bias built into their formulae. We try to rectify that making use of degrees of freedom.

For instance $\hat{\sigma}^2$, (the estimator for true population σ^2) given by $\sum e_i^2 / df$ no longer remains unbiased. And this very $\hat{\sigma}^2$ enters into calculation of standard errors of OLS estimates.

- 5) Since, estimates of standard errors are themselves no longer reliable, we may end up drawing wrong conclusions using conventional reasoning based on procedures for testing the hypothesis.

How to detect Heteroscedasticity

In applied regression analysis, plotting the residual terms can give us important clues about whether or not one or more assumptions underlying our regression model hold. The pattern exhibited by e_i^2 plotted against the concerned variable can provide important clue. If no pattern is detected – homoscedasticity holds or heteroscedasticity is absent. On the other hand, if errors form a pattern with variable — expanding, increasing linear or changing in some non-linear manner thus heteroscedasticity is certainly present.

Some tests have been designed to detect presence of Heteroscedasticity, using various statistical techniques. Prominent ones are: Park Test, Glejser Test, Whites General Test, Spearman's Rank correlation Test, Goldfield - Quandt Test etc. But in this unit, the limitation of space does not permit us to go into their details. We are forced to refer the learners again the course on Econometric Method for details in this regard.

How to tackle the heteroscedasticity?

Our ability to tackle the problem will depend upon the assumptions. We can really make about error variance. Thus, the following situations may emerge

- i) When σ_1^2 is known

Here the CLRM

$Y_i = \beta_0 + \beta_1 X_i + u_i$ can be transformed, dividing each value by corresponding σ_1 thus,

$$\frac{Y_i}{\sigma_1} = \beta_0 \left(\frac{1}{\sigma_1} \right) + \beta_1 \left(\frac{X_i}{\sigma_1} \right) + \frac{U_i}{\sigma_1}$$

This effectively transforms error terms to $U_i = \frac{U_i}{\sigma_1}$ which is homoscedastic

and therefore, the OLS estimators will be free of disability caused by heteroscedasticity. The estimates of β_0 and β_1 in this situation are called **Weighted Least Squares Estimators (WLSEs)**.

- ii) **When σ^2 is unknown:** we make some further assumptions about error variance:
- iiia) Error variance proportional to the X_i s. Here, the Square Root transformation is enough. We divide on both sides by $\sqrt{X_i}$. Thus, our regression line looks like:

$$\frac{Y_i}{\sqrt{X_i}} = \frac{\beta_0}{\sqrt{X_i}} + \beta_1 \frac{X_i}{\sqrt{X_i}} + \frac{U_i}{\sqrt{X_i}}$$

$$= \beta_0 \frac{1}{\sqrt{X_i}} + \beta_1 \sqrt{X_i} + U_i$$

Here $U_i = \frac{U_i}{\sqrt{X_i}}$ and this is sufficient to address the problem.

ii) Error Variance proportional to X_i^2 . Here, instead of division by $\sqrt{X_i}$, we divide by X_i on both the sides and estimate

$$\begin{aligned} \frac{Y_i}{X_i} &= \beta_0 \frac{1}{X_i} + \beta_1 + \frac{U_i}{X_i} \\ &= \beta_0 \frac{1}{X_i} + \beta_1 + U_i \end{aligned}$$

The error term will be $U_i = \frac{U_i}{X_i}$ and this will be free of heteroscedasticity, facilitating use of CLS techniques.

iii) **Respecification of Model:** Assigning a different functional form to the model, in place of speculating about the nature of variance may be found expedient. We can estimate this model:

$$\ln Y_i = \beta_0 + \beta_1 \ln X_i + U_i$$

This loglinear model is usually adequate to address our concerns.

10.8 PROBLEM OF AUTOCORRELATION

The classical regression model also assumes that disturbance terms U_i s do not have any serial correlation. But, in many situations this assumption may not hold. The consequences of the presence of serial or autocorrelation are similar to those of heteroscedasticity: the OLS are no longer BLUE.

Symbolically **no** autocorrelation means $E(U_i U_j) = 0$ when $i \neq j$. Autocorrelation can arise in economic data on account of many factors:

- i) there may be various reasons which cause cyclical up and down swings in economic time series. These tendencies continue till something happens which reverse them. This is called **inertia**.
- ii) Misspecification of model in the form of under specification can be a cause of autocorrelation: you have fewer X variables in the model, leaving out rather large systematic components to be clubbed with errors.
- iii) Cob-web phenomenon is another factor which may create this problem in certain types of economic time series (especially agricultural output etc.).
- iv) **Polishing of data** – like adding monthly data to make quarterly or quarterly to **make** half yearly series etc. can also be responsible for the autocorrelation – as the averaging involved dampens the fluctuations of the original data.

The consequences of autocorrelation are not different from those of heteroscedasticity listed in 10.7 above. Here too OLS estimators are biased or are not BLUE, t & F tests are no longer reliable. Therefore, computed value of R^2 is not reliable estimate of true goodness of fit.

There are many tests for detecting autocorrelation – varying from visual inspection of error plots, the Runs Test, Swed-Eisenhart critical runs test. But most commonly used is Durbin-Watson d test defined as

$$d = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$$

However, again, we are holding back information on practical detections and avoidance of problem of autocorrelation for the reasons of limitation of space here.

10.9 THE MAXIMUM LIKELIHOOD ESTIMATIONS

Sometimes another class of estimators is used in place of OLS. This class of estimators is called *maximum likelihood estimators* (MLEs). The MLEs possess some stronger theoretical properties. But it also requires a stronger assumption about distribution of error terms. Moreover, when errors follow normal distribution, we find that OLS and MLE methods give identical estimates of β parameters, both in simple and in multiple regressions. However, MLE estimate of σ^2 is biased. Hence, if one uses assumption of normal distribution of U_i s and persists with OLS, one does not miss out on any thing that may be advantageous in MLE.

The Method: Simple Regression Illustration

In the model $Y_i = \beta_0 + \beta_1 X_i + U_i$, the Y_i is normally and independently distributed with means $\beta_0 + \beta_1 X_i$ and variance σ^2 . Therefore, the joint probability density function of

$Y_1, Y_2, Y_3, \dots, Y_n$ with above mean and variance will be

$$f(Y_1, \dots, Y_n / \beta_0 + \beta_1 X_i, \sigma^2) \quad \text{ML - 1}$$

But given the independence of Y_i s, this function can be written as product of ' n ' individual density functions, or

$$\begin{aligned} f(Y_1, \dots, Y_n / \beta_0 + \beta_1 X_i + U_i, \sigma^2) \\ = \pi [Y_i / \beta_0 + \beta_1 X_i + U_i, \sigma^2] \end{aligned} \quad \text{ML-2}$$

$$\text{where } f(Y_i) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\left[-\frac{1}{2} \frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{\sigma^2} \right]} \quad \text{ML-3}$$

Putting this value for each Y_i , in ML-2. We get the likelihood function (LF):

$$LF(\beta_0, \beta_1, \sigma^2) = \frac{1}{\sigma^n (\sqrt{2\pi})^n} \cdot e^{\left[-\frac{1}{2} \sum \frac{(Y_i - \beta_0 - \beta_1 X_i - U_i)^2}{\sigma^2} \right]} \quad \text{ML - 4}$$

The method of maximum likelihood estimation is nothing but maximization of the (LF) given in ML-4 above. We can differentiate logarithms of (LF) with respect to β_0 , β_1 and σ^2 and equate the respective partials to zero to get the requisite estimation relations. So as

$$\ln(LF) = n \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum \frac{(Y_i - \beta_0 - \beta_1 X_i - U_i)^2}{\sigma^2} \quad \text{ML - 5}$$

therefore

$$\frac{\partial \ln(LF)}{\partial \beta_0} = \frac{-1}{\sigma^2} \sum (Y_i - \beta_0 - \beta_1 X_i)(-1) = 0 \quad \text{ML - 6}$$

$$\frac{\partial \ln(LF)}{\partial \beta_1} = \frac{-1}{\sigma^2} \sum (Y_i - \beta_0 - \beta_1 X_i)(-X_i) = 0 \quad \text{ML - 7}$$

$$\frac{\partial \ln(LF)}{\partial \sigma^2} = \frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (Y_i - \beta_0 - \beta_1 X_i)^2 = 0 \quad \text{ML - 8}$$

Simplification of ML-6 and ML-7 give us

$$\sum Y_i = n\beta_0 + \beta_1 \sum X_i \quad \text{ML-8a}$$

$$\sum X_i Y_i = \beta_0 \sum X_i + \beta_1 \sum X_i^2 \quad \text{ML-9}$$

Which are same as OLS normal equations.

Substituting values of ML (=OLS) estimates obtained by simultaneously solving ML-8 and ML-9 into ML-7 gives us

$$\begin{aligned} \sigma^2 &= \frac{1}{n} \sum (Y_i - \beta_0 - \beta_1 X_i)^2 \\ &= \frac{1}{n} \sum U_i^2 \end{aligned} \quad \text{ML - 10}$$

$$\text{Expectation of } \tilde{\sigma}^2, E(\tilde{\sigma}^2) = \frac{1}{n} E(\sum U_i^2)$$

$$= \left(\frac{n-2}{n} \right) \sigma^2$$

$$\sigma^2 - \frac{2}{n} \sigma^2$$

or $\tilde{\sigma}^2$ is biased downwards.

Multiple Regression: An example: The following regression were run on SPSS. Edited summary of results is as follows:

All figures are in Millions of US dollars

Year	var00001	var00002	var00003
1991-1992	19411.00	17865.00	133.00
1992-1993	21882.00	18537.00	559.00
1993-1994	23306.00	22238.00	4153.00
1994-1995	28654.00	26330.00	5138.00
1995-1996	36678.00	31797.00	4892.00
1996-1997	39133.00	33470.00	6133.00
1997-1998	41484.00	35006.00	5385.00
1998-1999	42389.00	33218.00	2401.00
1999-2000	49671.00	36822.00	5181.00
2000-2001	50536.00	44560.00	5862.00
2001-2002	51413.00	43827.00	6693.00
2002-2003	61412.00	52719.00	4555.00

var00001=Imports

var00002=Exports

var00003=Foreign Investment Inflow

Regression of Imports on Exports and Foreign Investment Inflow

R	R ²	$\bar{R}^2$	Standard error of Estimate
0.983	0.966	0.958	2726.228

Coefficients

	B	Standard Error	t
Constant	-1655.737	2658.578	-0.623
var00002	1.263	0.104	12.192
var00003	-0.288	0.522	-0.552

Coefficients Correlation

Correlation		Var00003	Var00002
	Var00003	1.0000	-0.670
	Var00002	-0.670	1.0000
Covariance	Var00003	0.272	-3.625E-02
	Var00002	-3.625E-02	1.073E-02

ANOVA

Sum of Sqrs.	df	Mean sqrs.	F
Regression 1.9E+09	2	9.4E+08	127.097
Residual 6.7E+07	9	7432320	
Total 2.0E+09	11		

Regression of Foreign Investment Inflow on Exports and Imports

R	R ²	$\bar{R}^2$	Standard error of Estimate
0.684	0.468	0.349	1712.334

Coefficients

	B	Standard Error	t
Constant	-321.545	1702.075	-0.189
var00001	-0.114	0.206	-0.552
var00002	0.272	0.257	1.060

Coefficients Correlation

Correlation		Var00002	Var00001
Covariance	Var00002	1.0000	-0.982
	Var00001	-0.982	1.0000
	Var00002	6.590E-02	-5.92E-02
	Var00001	-5.192E-02	4.24E-02

ANOVA

	Sum of Sqrs.	df	Mean sqrs.	F
Regression	2.3E+07	2	1.2E+07	3.952
Residual	2.6E+07	9	2932089	
Total	5.0E+07	11		

The above regressions are based on certain 'a priori' expectations. We expect that imports into India during the period 1991-92 to 2002-03 depend upon exports from India and foreign investment inflows into the country. The idea is that exports pay for imports and foreign investment inflow 'facilitates' the country to import more. Our results confirm our theoretic expectations. Regression on imports on exports and investment inflows shows that $R^2 = 0.966$ and $\bar{R}^2 = 0.958$. This shows that model has high explanatory power. It explains over 95 per cent of the variation. Typical computer output gives information on coefficients, correlation and co-variances, co-linearity diagnostics, residual analysis of variance etc. We have run another regression too – the results of which are reported above. This is regression of foreign investment

inflows on exports and imports. Theoretic expectation was that investment inflows are determined by magnitudes of exports and imports. However, this model gives rather disappointing results. $R^2 = 0.468$ and $\overline{R}^2 = 0.349$ only. In other words, our model explains less than 35 per cent of the variations. It is not desirable to persist with it. The same is reflected in low values of β_1 & β_2 parameters and their high standard errors.

Check Your Progress 3

- 1) What is hetero-scedasticity? What are its consequences?

.....

.....

.....

.....

.....

.....

- 2) What is auto-correlation? When does it arise? What are its consequences?

.....

.....

.....

.....

.....

.....

- 3) What are maximum likelihood estimations? Why do we persist with least squares estimators most of the time.

.....

.....

.....

.....

.....

.....

10.10 LET US SUM UP

We began with extension of the simple linear regression model to incorporate one more explanatory variable. Our next step was to interpret the coefficients of partial regression. Afterwards, we tried to design statistical touch stone for

inclusion of more variables into the model. This also gave us some guidelines to drop the ‘undesirable’ variable as well. We then moved on to more tedious and mathematically more demanding extension to ‘n’ variable model. We then attempted to analyse the effects of multi-co-linearity, heteroscedasticity and auto-correlation respectively. We also made comments on techniques of identification of these problems and some strategies to get rid of the problems as well. Further, we discussed the class of estimators called maximum likelihood estimators (MLEs). We made comparison between MLEs and Least square estimators as well. Finally, we ran some regressions on SPSS package between India’s imports, exports and foreign investment inflows to illustrate some of the points, which arose in course of our discussions. However, we must again draw the attention of learners to the fact that limitations of space did not permit us to go in for an exhaustive discussion of the concepts touched upon. Those who want to have detailed knowledge about the issues and concepts involved in multiple regression models are advised to go through our optional course MECE-001 Econometric Methods.

10.11 KEY WORDS

Multiple Regressions	: Regression of one dependent variable on more than one independent variables.
Partial regression Coefficients	: Coefficients of individual predictors in multiple regressions.
Coefficient of multiple determination — R^2	: Ratio of variation explained or accounted for by the regression model to the total variation.
Adjusted coefficient of multiple determination — $\bar{R}^2$	: It is coefficient of determination adjusted for the loss of degrees of freedom. It helps us against increasing $\bar{R}^2$ by incorporating unnecessary explanatory variables.
Multi-co-linearity	: Existence of linear relationships between explanatory variables in addition to one between dependent variable and them.
Heteroscedasticity	: Differences between variances of the error terms. This problem is more serious in cross-sectional data.
Auto-co-relation	: Correlations between the errors and their previous values. This problem is encountered very often when dealing with time series data.

10.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

- 1) Read Section 10.2
- 2) Read Section 10.2
- 3) Read Section 10.2

- 4) Read Section 10.3
- 5) Read Section 10.4
- 6) Read Section 10.4
- 7) Read Section 10.4
- 8) Read Section 10.6
- 9) Read Section 10.7
- 10) Read Section 10.8
- 11) Read Section 10.9

